United International **University (UIU)**

Dept. of Computer Science & Engineering (CSE)

Final Exam: : Trimester: Spring 2026

Course Code: CSE 1111, Course Title: STRUCTURED PROGRAMMING LANGUAGE

Time: 2 hours   Total Marks: 40

(Unfair means in the exam will be taken seriously as per UIU disciplinary rules)

Answer all the Questions. Write your serial number of exam attendance sheet on your answer script.

1 a) Show **manual tracing** for the following program and find **output**. 5

```
#include<stdio.h>
int a, b;
float func1(int x);
void func2(float x, int y);
int main(){
      a=18; b=22;
      printf("%f\n", a+func1(a+60)+b);
      func2(6.5, 20);
      printf("%f\n", b+func1(a+70)+a);
      return 0;
 }
 float func1(int x) {
      a=x*10;
      printf("%d\n", x+35);
      b=a+b;
      return x+a+b+10.5;
 }
void func2(float x, int y){
      printf("%f\n", x+5);
      a=a+y;
      b=b+x+y;
 }
```

b) Write a **program** using **two user defined functions** to find the sum of the following series 5

$$2^3-4^3+6^3-8^3+\ldots\ldots-100^3$$

Here,

i) The function **main()** calls the user defined function **Evaluation(int a, int b)** to find the sum of the above-mentioned series and to return the sum**.** Here, the passing values from **main()** to the **Evaluation()** are 2 and 100, respectively.

ii) The user defined function **Evaluation(int a, int b)** calls **TermValue(int x, int y)** to calculate x^y, where x starts at **a** and ends at **b**.

iii) The **main()** function prints the returned value from the function **Evaluation()**.

2 a) Show **manual tracing** for the following program 4

```
char Str1[13]={'\0'};
char Str2[4]={'\0'};
int i=0, j, k=0;
strcpy(Str1, "CSE DEPT ");
strcpy(Str2, "UIU");
for(; Str1[i++]!='\0' ;)
printf("%s  %s  %d\n", Str1, Str2, i);
for(j=i; j<i+strlen(Str2); j++){
   Str1[[j]=Str2[k++];
}
printf("%s  %s  %d\n", Str1, Str2, j);
```

```
if (strcmp(Str1, Str2)<0)
  printf("String %s is smaller", Str2);
else
  printf("String %s is greater", Str1);
```

b) Distinguish between call by value and call by reference. 2

c) Show **manual tracing** for the following program and find **output**. 2

```
#include<stdio.h>

void func(int a, int *b){
  int *q;
  q=b;
  *q=*q+a;
  a=a-4;
}
int main(){
  int a=10, b=19;
  printf("%d  %d\n", a+b, b-a);
  func(a, &b);
  printf("%d   %d", a+b, b-a);
  return 0;
}
```

3 a) Show **manual tracing** for the following program and find **output**. 5

```
#include<stdio.h>
void func(int *B, int n);
int main(){
  int A[5]={15, 25, 35, 45, 50};
  int i;
  for(i=0; i<5; i++)
    printf("%d  ", A[i]);
  func(A, 5);
  printf("\n");
  for(i=0; i<5; i++)
    printf("%d  ", A[i]);
  return 0;
}
void func(int *B, int n){
  int i;
  for(i=0; i<n; i++){
       *(B+i)=*(B+i)+i;
  }
}
```

b) i) Distinguish between write and append modes in file writing. 2

ii) What type of problems will be happened if a file is not closed after opening? 2

iii) Write a program that reads two floating point numbers from a file and write the sum of the numbers in the same file. 2

4 Write a **program** having the structure **student (name, id, course name, phone, email)** to perform the following operations for n students 7

a) Read name, id, course name, phone and email of n students from keyboard.
b) Sort the students in ascending order of id.
c) Display the following sample report on monitor for the three students (n=3)

Rahim	10	SPL	01632078812	xxx@gmail.com
Saiham	20	ICS	01710012300	yyy@hotmail.com
Sabiha	25	DSA-I	01893476433	zzz@yahoo.com

5 Write a **program** to find the total number of spaces, capital letters and small letters in an input string. 4

For example,
Input String: "Where iS YoUR   TowN"
Output:

Spaces=5
Cap_letters=7
Small_letters=8